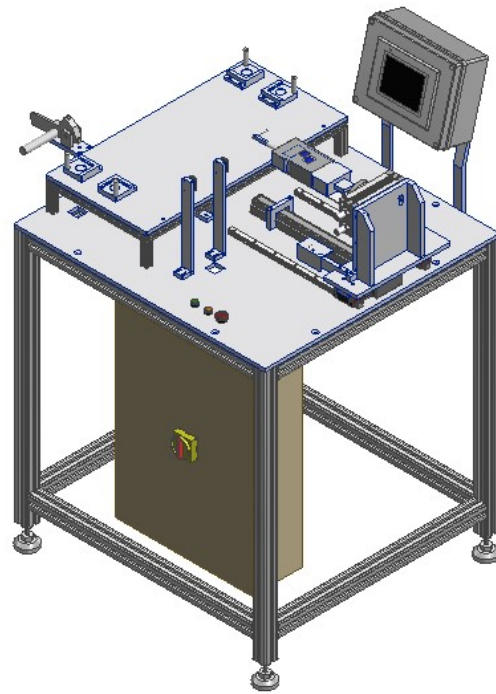
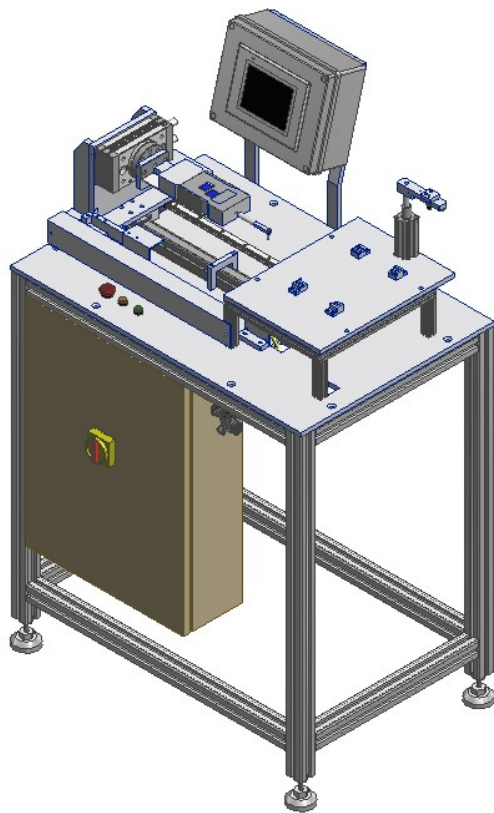
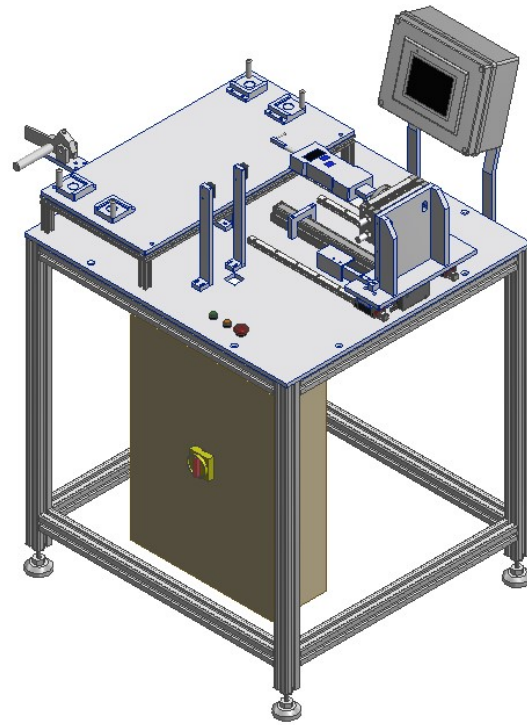




X BELT TENSION ADJUST MACHINE

USER MANUAL



MACHINE INTRODUCTION

This machine is an X-Belt Tensioner Adjustment Machine.

There are 3 type of machines,

Cupid , Summer and Voldemort.

To simplify adjustment, real time tensioner reading are display at the LCD Monitor.

OPERATION MANUAL

MACHINE OPERATION PROCEDURE

1. TURN ON THE IMADA (FORCE GAUGE).
2. AFTER 10 SECONDS, POWER ON SWITCH AT CONTROL BOX AND MAKE SURE EMERGENCY BUTTON IS IN OFF POSITION.
3. PRESS RESET BUTTON AT TOUCH PANEL DISPLAY AND MACHINE WILL PERFORM HOMING ROUTINE.
4. SET MACHINE TO AUTO MODE.
5. PUT PRODUCT TO MACHINE FOLLOWING GUIDES POSITION AND PRESS THE START BUTTON.
6. WHEN PROCESS IS COMPLETED, OPERATOR CAN UNLOAD THE PRODUCT FROM THE MACHINE.

MACHINE FRONT PANEL

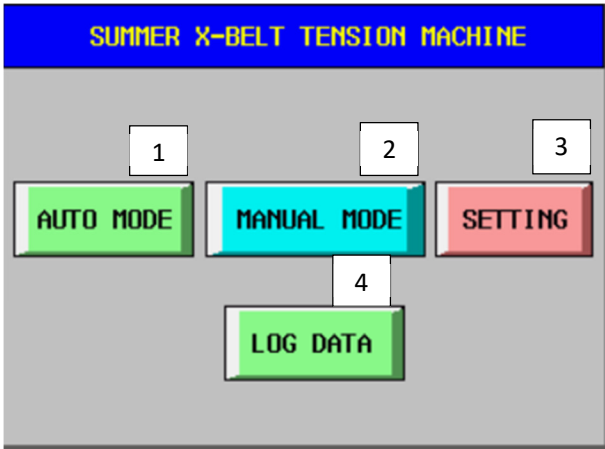


Figure 1: Main panel

MAIN PANEL

No	Function	Description
1	AUTO MODE	When turned on, the machine is in auto mode and ready for production. When turned off, the machine is in debugging or repairing mode.
2	MANUAL	TO enter IO & actuator setting mode.
3	SETTING	To enter setting mode.
4	LOG DATA	View the log data for this machine.

AUTO FUNCTION

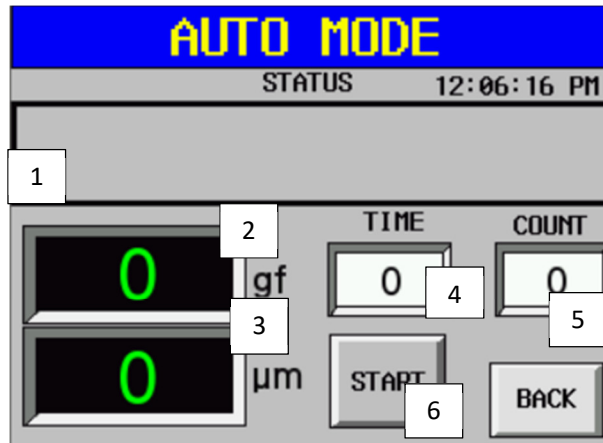


Figure 1: Auto Screen

AUTO SCREEN

No	Function	Description
1	STATUS	Machine status.
2	FORCE	Force reading from IMADA. Unit gram-force (gf).
3	DISTANCE	Distance reading from sensor. Unit micrometre (μm)
4	TIMER	Cycle time for full process
5	COUNT	Product counter.
6	START	Start process without using carriage sensor.

SETTING FUNCTION

SETTING 1				SETTING 2					
ZERO FORCE	MIN 0	MAX 0	gf	QC VALUE	CALIBRATION SPEC	MIN 0	MAX 0	μm	USB OFF
ADJUSTMENT FORCE	0	0	gf	NORMAL MODE	CALIBRATION FORCE	0	gf		
OFFSET	-	+	gf	PRODUCTION MODE					
CARRIAGE CYCLE	0								
RESET COUNTER				NEXT	BACK	CALIBRATION PAGE		SUMMER	BACK

Figure 2: Setting Screen

SETTING SCREEN

No	Buttons	Description
1	ZERO FORCE	Initial contact force value setting
2	ADJUSTMENT FORCE	5mm push force value setting
3	OFFSET	Offset value for adjustment force. (QC Mode only)
4	CARRIAGE CYCLE	Carriage movement counts in starting process.
5	QC VALUE	QC Value for adjustment force.
6	NORMAL MODE / SKIP STEP	Enable carriage movement in the middle of process / Disable carriage movement in the middle of process
7	PRODUCTION / QC MODE	Enable or disable bottom clamp in auto.

MANUAL FUNCTION

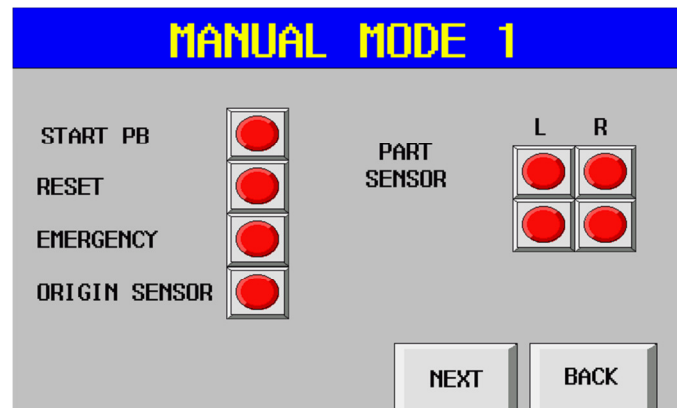


Figure 3: Input Screen

INPUT SCREEN

No	Function	Description
1	INPUT	To check the sensor/input signal.

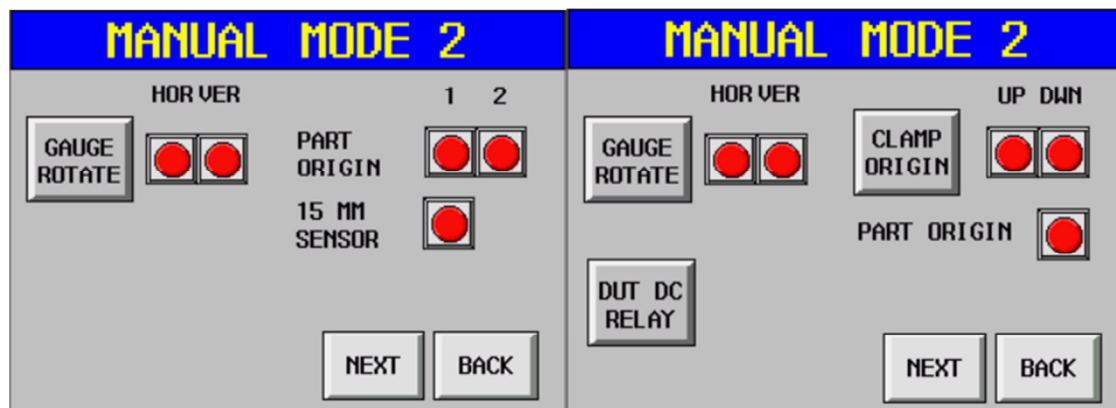


Figure 4: Output Screen

OUTPUT SCREEN

No	Function	Description
1	OUTPUT	Manual trigger output.

ACTUATOR SETTINGS

The screenshot displays the 'ACTUATOR SETTINGS' interface, divided into two main sections: 'MANUAL MODE 3' and 'MANUAL MODE 4'.

MANUAL MODE 3 (ACTUATOR):

- Navigation:** Includes left and right arrow buttons for jogging.
- Buttons:** 'ORIGIN' (red), 'MOVE' (yellow), 'FAST JOG' (white), 'SAVE' (pink), and 'GAUGE ROTATE' (grey).
- POS NO:** A numeric input field showing '1'.
- POS DESCRIPTION:** A list of positions:
 - 1 : STANDBY
 - 2 : PUSHER INSIDE
 - 3 : CLEARANCE POSITION
 - 4 : CALIBRATION POSITION
- Navigation:** 'NEXT' and 'BACK' buttons.

MANUAL MODE 4:

- Buttons:** 'START', '5MM PUSH', 'SLOW JOG', and 'FAST JOG'.
- ACC/DEC and SPEED:** Two columns of numeric input fields for acceleration/deceleration and speed settings, each with a '0' in the top field.
- Navigation:** 'BACK' button.

SMC I/O:

- Status Indicators:** Four red circular LEDs labeled 'SMC.BUSY', 'SMC.SETON', 'SMC.INP', and 'SMC.ALARM'.
- Buttons:** 'SMC.RESET' and 'BACK'.

Figure 5: Actuator Settings Screen

No	Function	Description
1	ARROW RIGHT	Right jog actuator.
2	ARROW LEFT	Left jog actuator.
3	ORIGIN	Move actuator to origin position.
4	FAST JOG	Enable or disable fast jog.
5	MOVE	Move actuator to selected position number.
6	SAVE	Save actuator position to selected position number.
7	ACC/DEC	Acceleration and deceleration setting for selected movement.
8	SPEED	Speed setting for selected movement.

LOG DATA

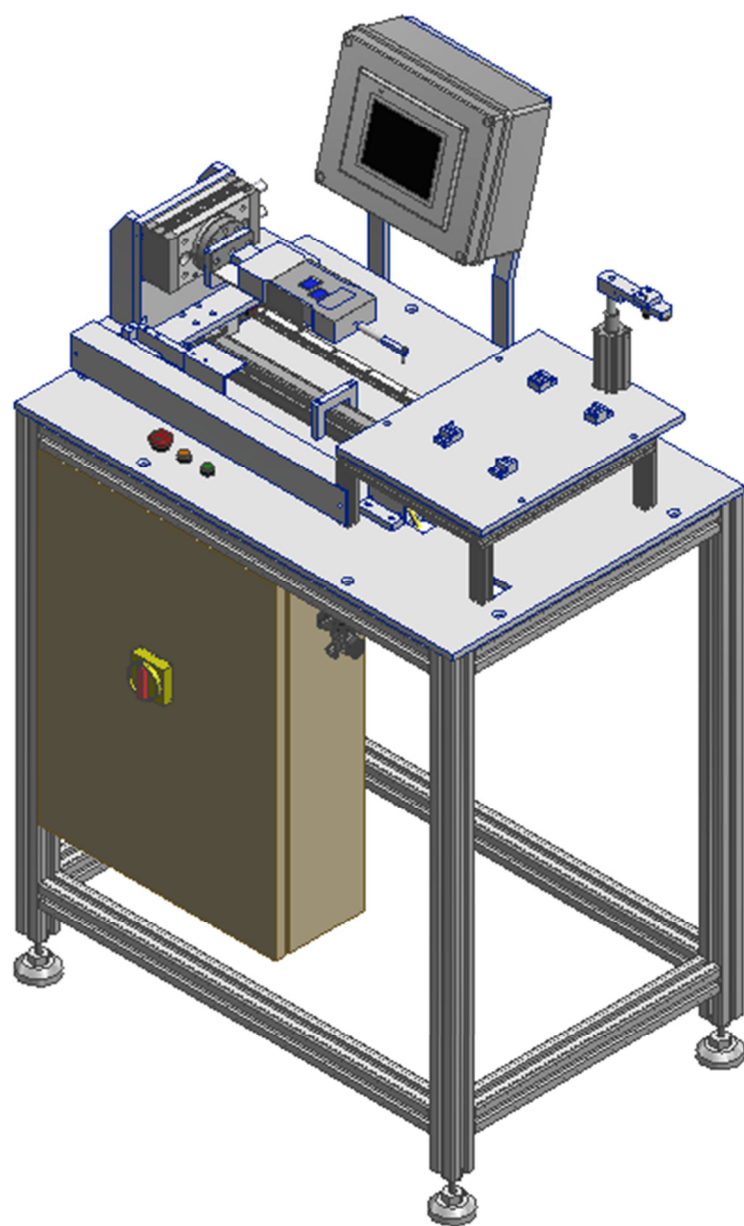
Date	Time	Reading (g)

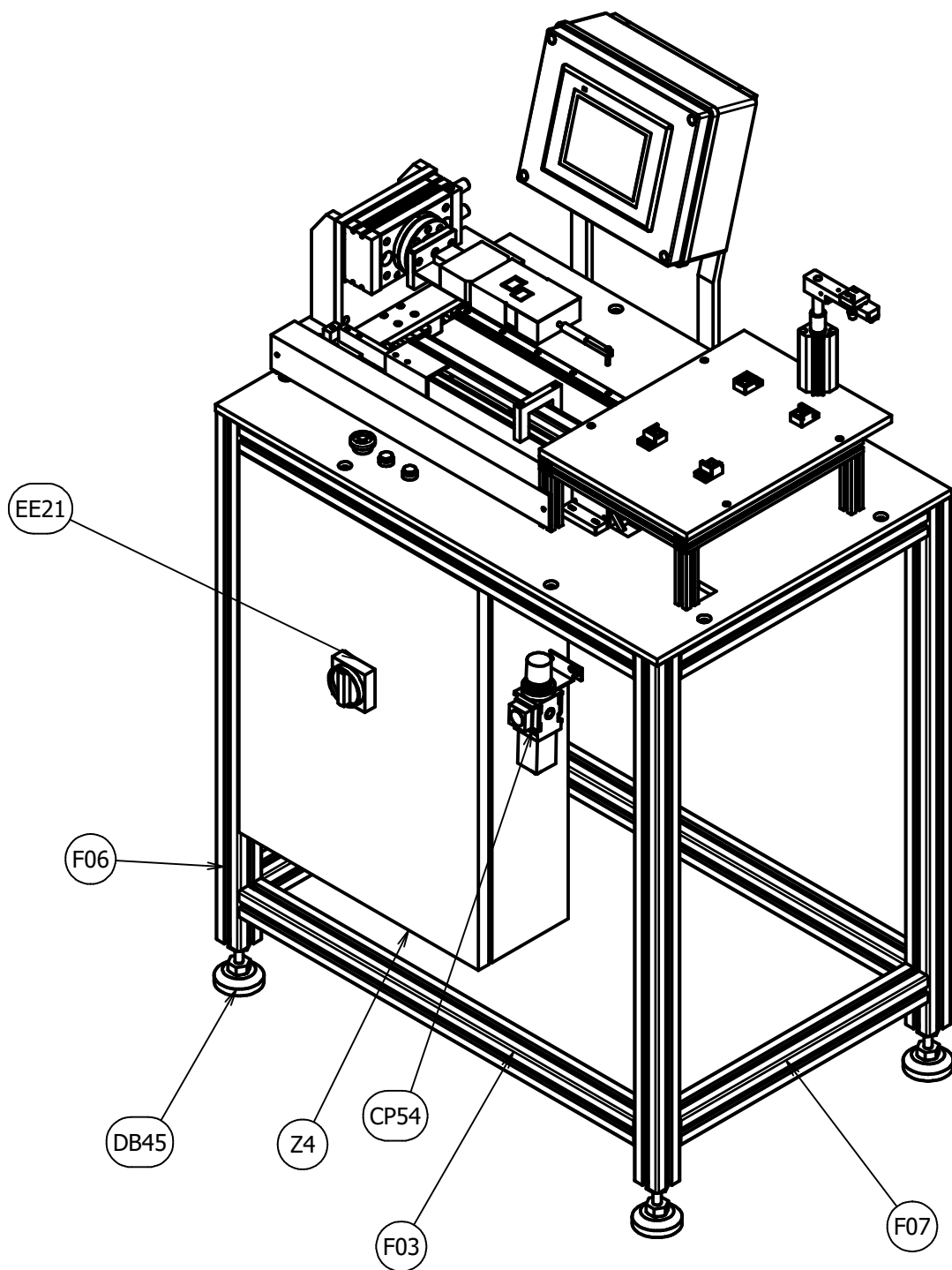
BACK

Figure 6: Log Data Screen

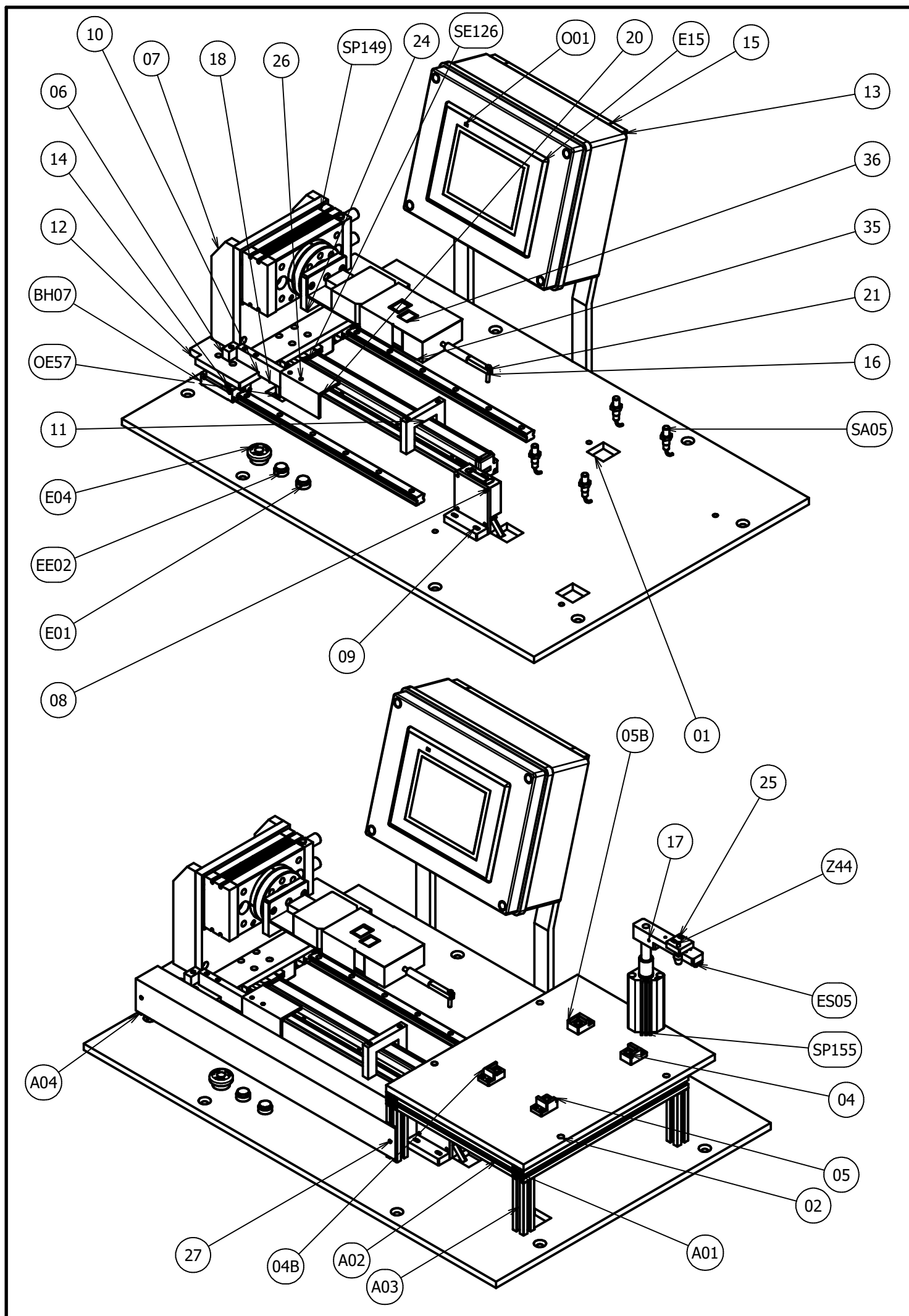
No	Function	Description
1	DATE	Testing data date.
2	TIME	Testing data time.
3	READING	Reading capture from tension gauge.

TYPE CUPID
MECHANICAL & ELECTRICAL
PART LIST



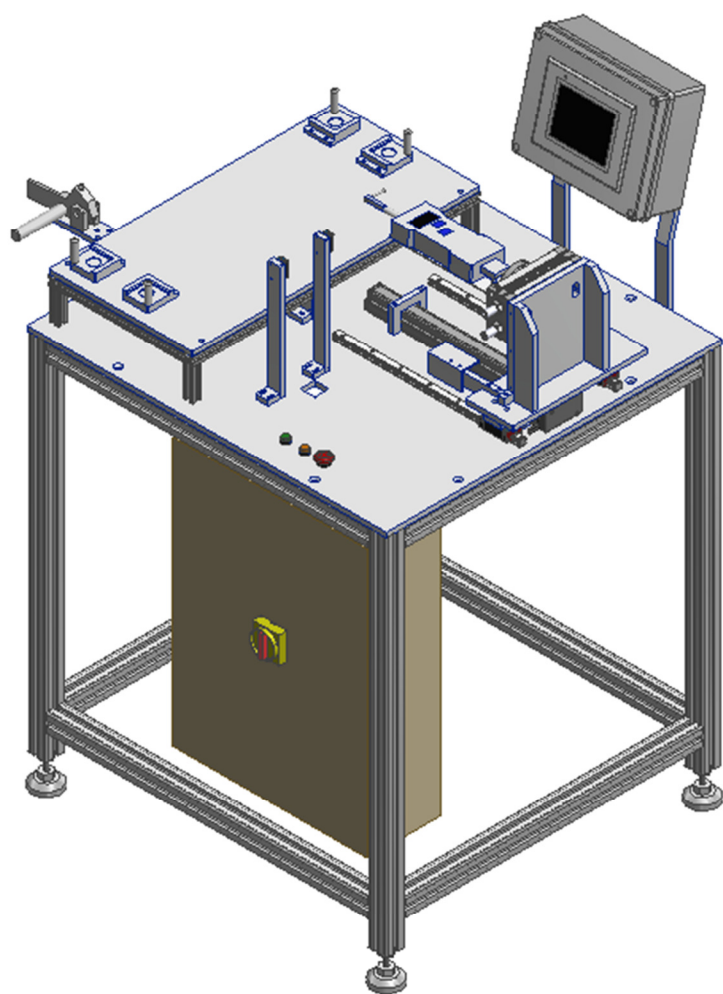


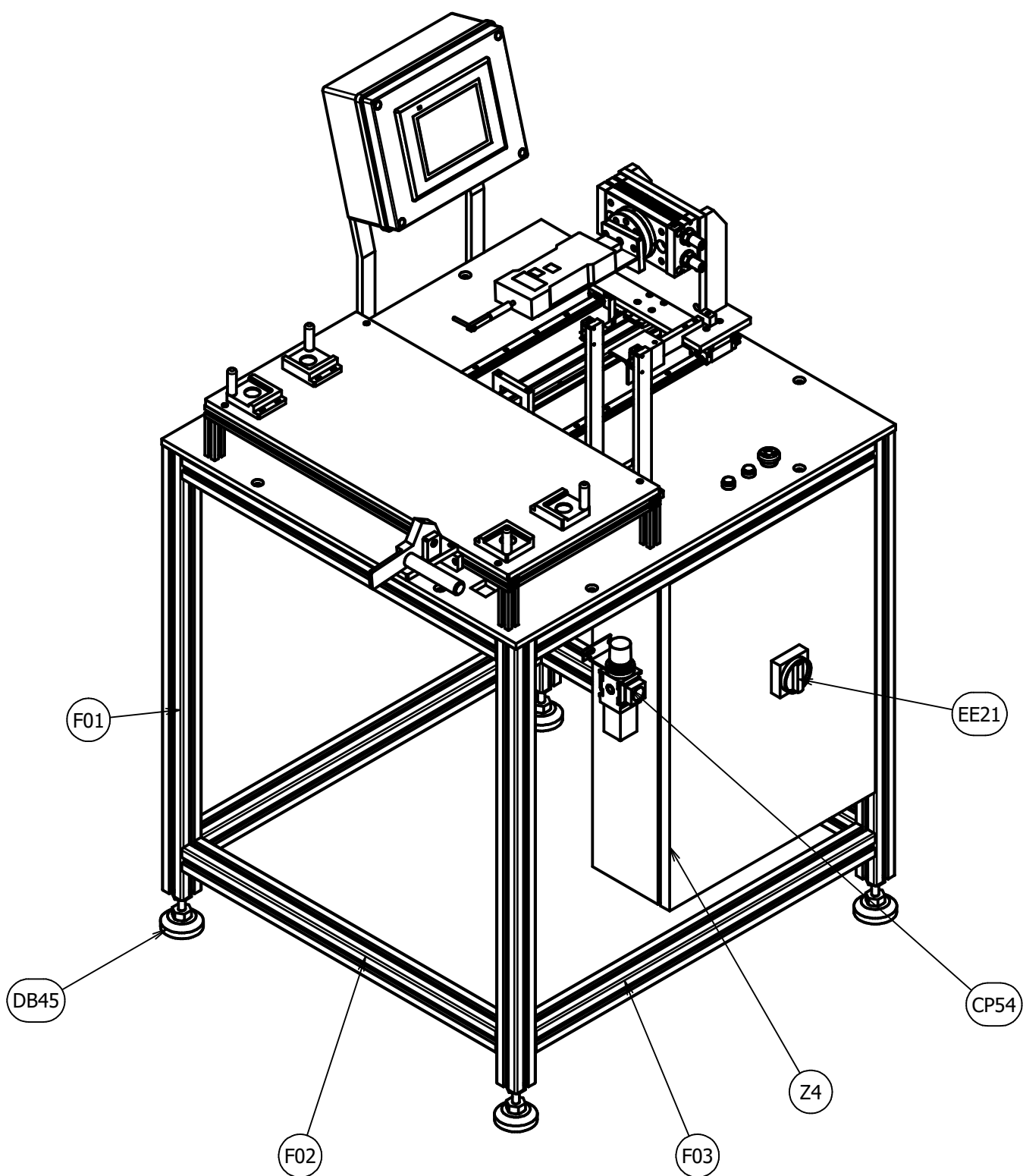
PARTS LIST	
ITEM No	PART NUMBER
CP54	W1000-8-W-B3W
DB45	FOOT CAP D60 M8
EE21	CKC-PS252
F03	AL 4040 3
F04	AL 4040 3 J
F06	AL 4040 1
F07	AL 4040 4 J
Z4	CBOX 6 x 16 x 22



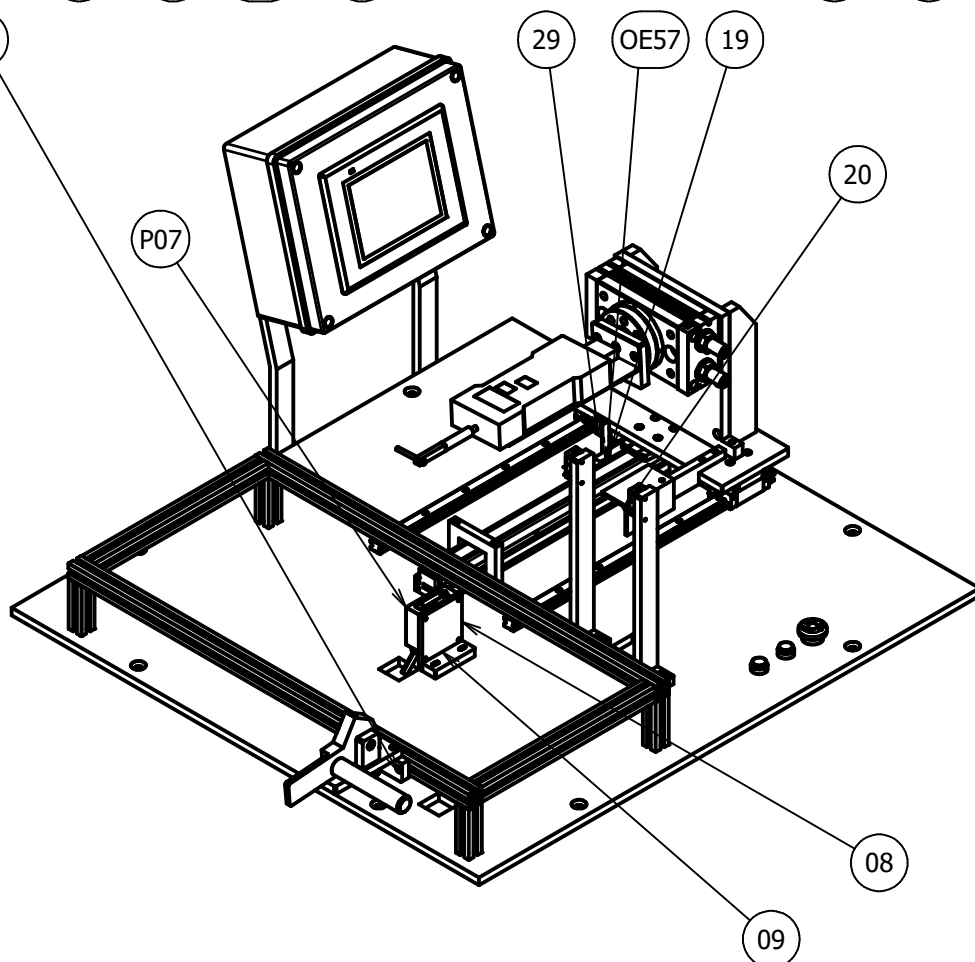
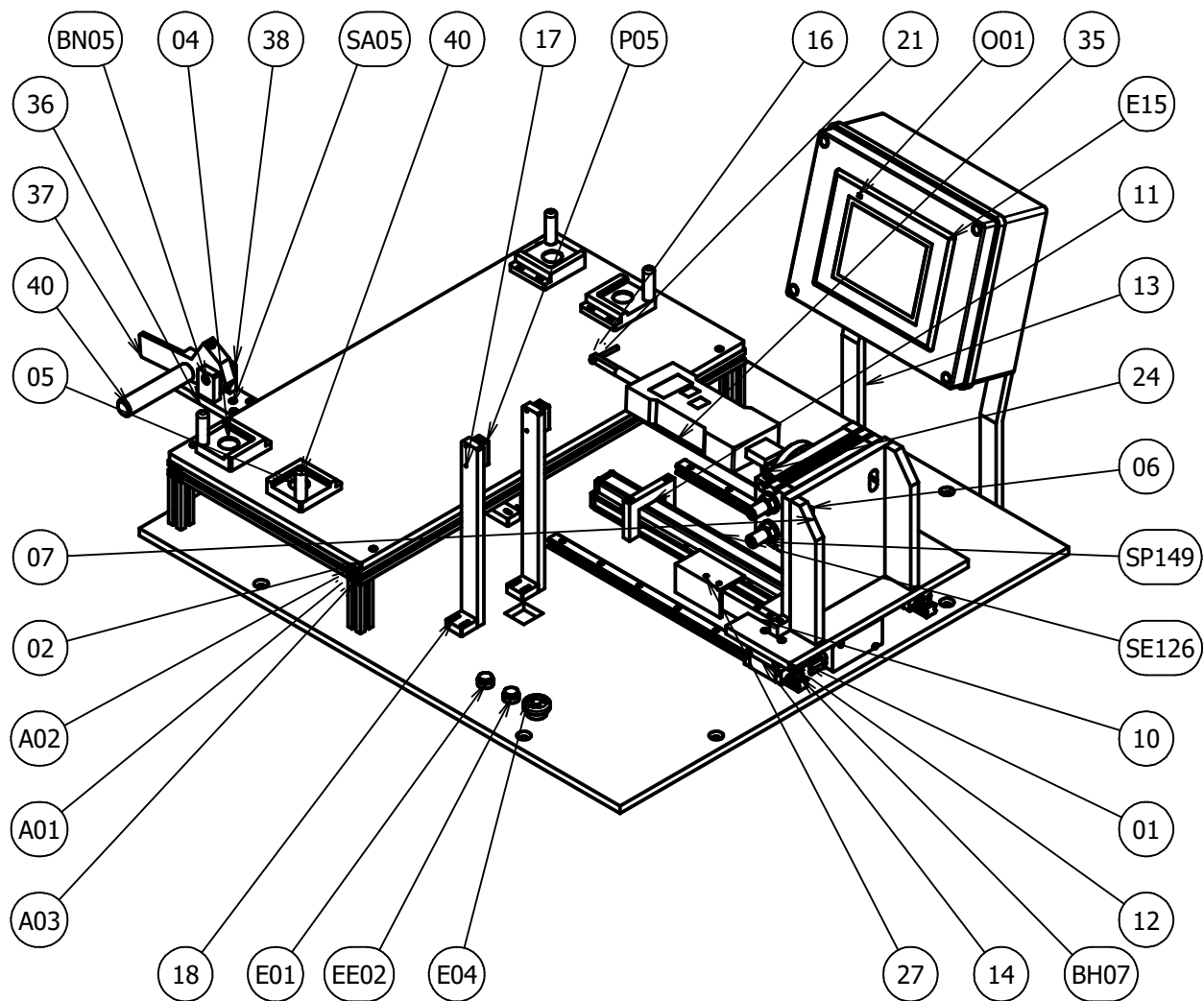
PARTS LIST	
ITEM No	PART NUMBER
01	TOP BASE CUPID
02	PART PLATE CUPID
04	G1J
04B	G1J B
05	G2J
05B	G2J B
06	STAND CYL 1
07	STAND CYL 2
08	D SENSOR PLATE
09	D SENSOR PLATE 2
10	SUS HOLDER C
11	S STOPPER
12	SLIDE PLATE JO
13	B STAND
14	LM BLOCK 16
15	B STAND 2
16	TEST PIN 1 J
17	SENSOR HOLDER
18	U SEN PLATE
19	U SENSOR L
20	R SENSOR PLATE
21	TEST PIN 2 CUPID
23	B1
24	B2
25	RUBBER PLATE
26	L COVER C
27	SAFETY L CUPID
35	CYL L PLATE A
36	SPRING PLATE
A01	AL 2020 1 C
A02	AL 2020 2 C
A03	AL 2020 3 C
A04	AL 2020 3 D
BH07	EGH15CAZ0H 390
E01	GREEN SW
E04	EMERGENCY SW S
E15	PLASTIC BOX 100 x 205 x 260 LCD
EE02	SRX-M25(Y)
ES05	BTF30-DDTL
O01	NB5Q TW00B
OE57	EE_SX672
SA05	PROXIMITY SENSOR M8
SE126	LEFS16LA-300-R3AN3
SP149	MSQB50A-M9B
SP155	MKB20-20RZ-M9B
Z44	RUBBER PRESS M4

TYPE SUMMER
MECHANICAL & ELECTRICAL
PART LIST



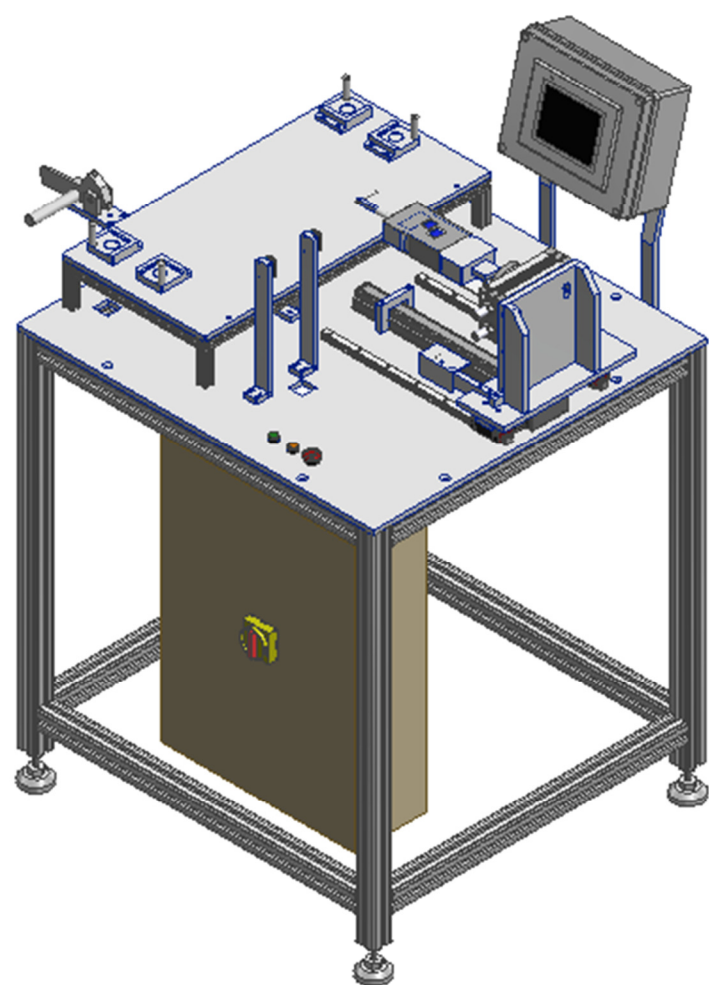


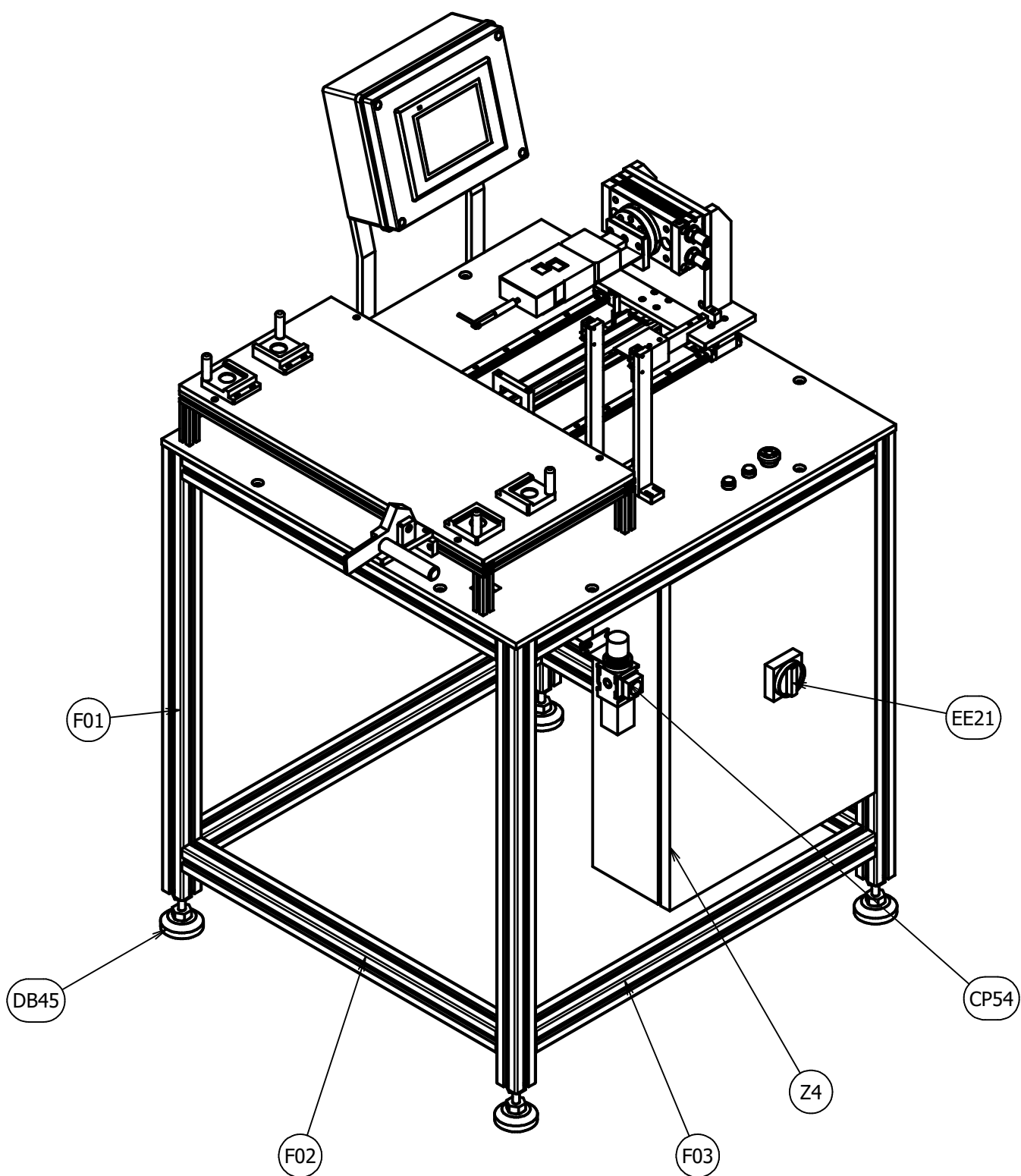
PARTS LIST	
ITEM No	PART NUMBER
CP54	W1000-8-W-B3W
DB45	FOOT CAP D60 M8
EE21	CKC-PS252
F01	AL 4040 1 S
F02	AL 4040 4
F03	AL 4040 3
Z4	CBOX 6 x 16 x 22



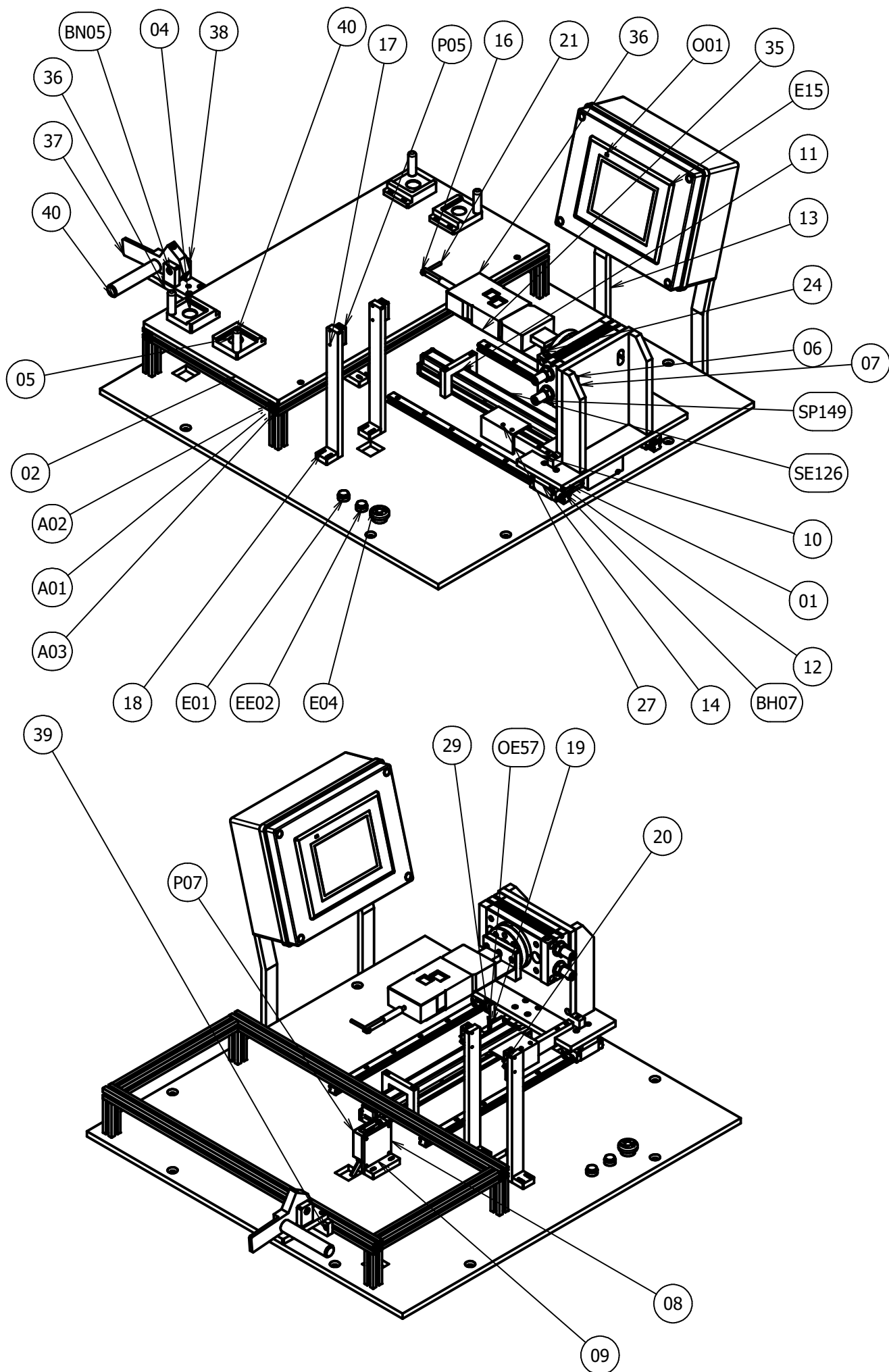
PARTS LIST	
ITEM No	PART NUMBER
01	TOP BASE SUMMER
02	PART PLATE SUMMER
04	G1S
05	G2B
06	STAND CYL 1 S
07	STAND CYL 2 S
08	D SENSOR PLATE
09	D SENSOR PLATE 2
10	SUS HOLDER S
11	S STOPPER
12	SLIDE PLATE SU
13	B STAND
14	LM BLOCK 16
15	B STAND 2
16	TEST PIN 1 S
17	BTL STAND 1
18	BTL STAND 2
19	U SENSOR L
20	R SENSOR PLATE
21	TEST PIN 2 S
23	B1
24	B2
27	L COVER S
29	U SEN PLATE
35	CYL L PLATE A
36	H PLATE
37	15 SLOT PLATE 1
38	H PLATE 2
39	H PLATE 3
40	G SHAFT 1
40	H SHAFT
A01	AL 2020 1 S
A02	AL 2020 2 S
A03	AL 2020 3 S
BH07	EGH15CAZ0H 390
BN05	606ZZ ID6 OD17 T6
E01	GREEN SW
E04	EMERGENCY SW S
E15	PLASTIC BOX 100 x 205 x 260 LCD
EE02	SRX-M25(Y)
O01	NB5Q TW00B
OE57	EE_SX672
P05	P1KH025
P07	hl-g103-a-c5
SA05	PROXIMITY SENSOR M8
SE126	LEFS16LA-300-R3AN3
SP149	MSQB50A-M9B

TYPE VOLDEMORT
MECHANICAL & ELECTRICAL
PART LIST



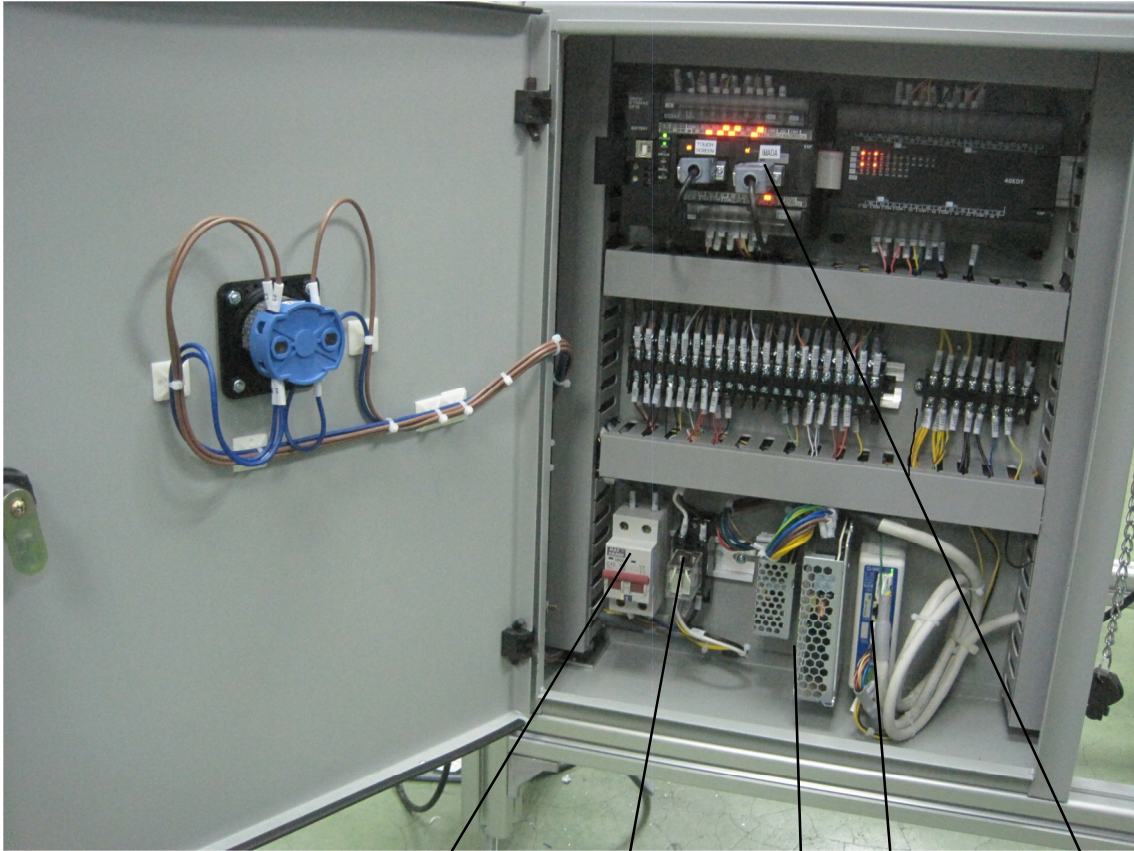


PARTS LIST	
ITEM No	PART NUMBER
CP54	W1000-8-W-B3W
DB45	FOOT CAP D60 M8
EE21	CKC-PS252
F01	AL 4040 1 S
F02	AL 4040 4
F03	AL 4040 3
Z4	CBOX 6 x 16 x 22



PARTS LIST	
ITEM No	PART NUMBER
01	TOP BASE VOLDEMORT
02	PART PLATE VOLDERMORT T1
04	G1S
05	G2B
06	STAND CYL 1 S
07	STAND CYL 2 S
08	D SENSOR PLATE
09	D SENSOR PLATE 2
10	SUS HOLDER S
11	S STOPPER
12	SLIDE PLATE SU
13	B STAND
14	LM BLOCK 16
15	B STAND 2
16	TEST PIN 1 S
17	BTL STAND 1
18	BTL STAND 2
19	U SENSOR L
20	R SENSOR PLATE
21	TEST PIN 2 S
23	B1
24	B2
27	L COVER S
29	U SEN PLATE
35	CYL L PLATE A
36	SPRING PLATE
36	H PLATE
37	15 SLOT PLATE 1
38	H PLATE 2
39	H PLATE 3
40	G SHAFT 1
40	H SHAFT
A01	AL 2020 1 S
A02	AL 2020 2 S
A03	AL 2020 3 S
BH07	EGH15CAZ0H 390
BN05	606ZZ ID6 OD17 T6
E01	GREEN SW
E04	EMERGENCY SW S
E15	PLASTIC BOX 100 x 205 x 260 LCD
EE02	SRX-M25(Y)
O01	NB5Q TW00B
OE57	EE_SX672
P05	P1KH025
P07	hl-g103-a-c5
SE126	LEFS16LA-300-R3AN3
SP149	MSQB50A-M9B

ALL TYPE COMMON
CONTROLLER BOX



1

2

3

4

5



6

7

ITEM	DESCRIPTION
1	MAGNETIC CIRCUIT BREAKER
2	RELAY
3	DC POWER SUPPLY
4	ACTUATOR CONTROLLER
5	PLC
6	3 PIN SOCKET
7	SOLENOID VALVE

PLC I/O AND ELECTRICAL
WIRING DIAGRAM
ALL TYPE

PLC I/O AND WIRING DIAGRAM TYPE CUPID

Model	PLC Mark	Label	Description	Remark	Device Pin
CP1E NA20DT-D	DC24V+	24Vdc	24V Power supply		24Vdc
	DC24 -	0Vdc	GND		GND
	E	E1	Earth		Earth
	COM	24Vdc	24V Power supply		24Vdc
CIO 0 (INPUT)	00	X000	START		
	01	X001	RESET		
	02	X002	EMERGENCY		
	03	X003	ACTUATOR ORIGIN		OMRON
	04	X004	PROXIMITY 1	} PART SENSOR	AUTONICS
	05	X005	PROXIMITY 2		AUTONICS
	06	X006	PROXIMITY 3		AUTONICS
	07	X007	PROXIMITY 4		AUTONICS
	08	X008	SMC.BUSY (Actuator Moving)		SMC Driver 1.CN5.14
	09	X009	SMC.SETON (Returned to Origin)		SMC Driver 1.CN5.15
	10	X010	SMC.INP (Position Complete)		SMC Driver 1.CN5.16
	11	X011	SMC.ALARM		SMC Driver 1.CN5.19
CIO 90 (INPUT)		V IN0	ANALOG VOLTAGE INPUT		WIRE BLACK
		I IN0			
		COM 0	ANALOG VOLTAGE GROUND		SHIELD
		AG			
CIO 91 (INPUT)		V IN1			
		I IN1			
		COM 1			
CIO 100 (OUTPUT)	00	Y10000	Stepper Driver 1	CW - Add 390 Ohm resistor	SMC Driver 1.CN5.6
	01	Y10001			
	02	Y10002	Stepper Driver 1	CWW- Add 390Ohm resistor	SMC Driver 1.CN5.4
	03	Y10003			
	04	Y10004	SMC.SETUP (HOME)		SMC Driver 1.CN5.7
	05	Y10005	SMC.CLR(RESET DEVIATION)		SMC Driver 1.CN5.10
	06	Y10006	SMC.RESET (reset Alarm)		SMC Driver 1.CN5.8
	07	Y10007	SMC.SVON (Servo Online)		SMC Driver 1.CN5.9
	COM	0Vdc	GND		0V
	COM	0Vdc	GND		0V
	COM	0Vdc	GND		0V
	COM	0Vdc	GND		0V
CIO 190 (OUTPUT)		VOUT0			
		I OUT0			
		COM 0			

CP1W-40EDT					
1CH (INPUT)	000	X100	UP	}	GAUGE ROTATE CYLINDER
	001	X101	DOWN		
	002	X102	UP	}	DUT ORIGIN CYLINDER
	003	X103	DOWN		
	004	X104	DUT ORIGIN SENSOR	PROX SPRING SENSOR	AUTONICS
	005	X105			
	006	X106			
	007	X107			
	008	X108			
	009	X109			
	010	X110			
	011	X111			
	COM	24Vdc	24V Power supply		24Vdc
2CH (INPUT)	000	X200			
	001	X201			
	002	X202			
	003	X203			
	004	X204			
	005	X205			
	006	X206			
	007	X207			
	008	X208			
	009	X209			
	010	X210			
	011	X211			
101CH (OUTPUT)	000	Y10100	UP	}	GAUGE ROTATE VALVE
	001	Y10101	DOWN		
	002	Y10102	UP	}	DUT ORIGIN VALVE
	003	Y10103	DOWN		
	004	Y10104	DUT DC RELAY (DC18V)		
	005	Y10105			
	006	Y10106			
	007	Y10107			
	COM	0Vdc	GND		0V
	COM	0Vdc	GND		0V
102CH (OUTPUT)	000	Y10200			
	001	Y10201			
	002	Y10202			
	003	Y10203			
	004	Y10204			
	005	Y10205			
	006	Y10206			
	007	Y10207			
	COM	0Vdc	GND		0V
	COM	0Vdc	GND		0V

SMC ACTUATOR Wiring

Model	SMC Ref	Label	Description	Remark	Device Pin
LECPAN3D CN1	BK RLS				
	EMG	24Vdc	24V Power supply		24Vdc
	C 24V	24Vdc	24V Power supply		24Vdc
	M 24V	24Vdc	24V Power supply		24Vdc
	0V	0Vdc	GND		0Vdc
LECPAN3D CN5	001	24Vdc	24V Power supply		24Vdc
	002	0Vdc	GND		0Vdc
	003	5Vdc	NP+		5Vdc
	005	5Vdc	PP+		5Vdc

PLC I/O AND WIRING DIAGRAM TYPE SUMMER

Model	PLC Mark	Label	Description	Remark	Device Pin
CP1E NA20DT-D	DC24V+	24Vdc	24V Power supply		24Vdc
	DC24 -	0Vdc	GND		GND
	E	E1	Earth		Earth
	COM	24Vdc	24V Power supply		24Vdc
CIO 0 (INPUT)	00	X000	START		
	01	X001	RESET		
	02	X002	EMERGENCY		
	03	X003	ACTUATOR ORIGIN		OMRON
	04	X004	PUSH BUTTON 1	} PART DETECT	
	05	X005	PUSH BUTTON 2		
	06	X006	PUSH BUTTON 3		
	07	X007	PUSH BUTTON 4		
	08	X008	SMC.BUSY (Actuator Moving)		SMC Driver 1.CN5.14
	09	X009	SMC.SETON (Returned to Origin)		SMC Driver 1.CN5.15
	10	X010	SMC.INP (Position Complete)		SMC Driver 1.CN5.16
	11	X011	SMC.ALARM		SMC Driver 1.CN5.19
CIO 90 (INPUT)		V IN0	ANALOG VOLTAGE INPUT		WIRE BLACK
		I IN0			
		COM 0	ANALOG VOLTAGE GROUND		SHIELD
		AG			
CIO 91 (INPUT)		V IN1			
		I IN1			
		COM 1			
CIO 100 (OUTPUT)	00	Y10000	Stepper Driver 1	CW - Add 390 Ohm resistor	SMC Driver 1.CN5.6
	01	Y10001			
	02	Y10002	Stepper Driver 1	CWW - Add 390Ohm resistor	SMC Driver 1.CN5.4
	03	Y10003			
	04	Y10004	SMC.SETUP (HOME)		SMC Driver 1.CN5.7
	05	Y10005	SMC.CLR(RESET DEVIATION)		SMC Driver 1.CN5.10
	06	Y10006	SMC.RESET (reset Alarm)		SMC Driver 1.CN5.8
	07	Y10007	SMC.SVON (Servo Online)		SMC Driver 1.CN5.9
	COM	0Vdc	GND		0V
	COM	0Vdc	GND		0V
	COM	0Vdc	GND		0V
	COM	0Vdc	GND		0V
CIO 190 (OUTPUT)		V OUT0			
		I OUT0			
		COM 0			

CP1W-40EDT					
1CH (INPUT)	000	X100	UP	} GAUGE ROTATE CYLINDER	
	001	X101	DOWN		
	002	X102			
	003	X103			
	004	X104			
	005	X105			
	006	X106			
	007	X107			
	008	X108			
	009	X109			
	010	X110			
	011	X111			
	COM	24Vdc	24V Power supply		24Vdc
2CH (INPUT)	000	X200	DUT SENSOR 1	} ORIGIN DUT MOTOR	OMRON
	001	X201	DUT SENSOR 2		OMRON
	002	X202	15MM GAP SENSOR	PROXIMITY	AUTONICS
	003	X203			
	004	X204			
	005	X205			
	006	X206			
	007	X207			
	008	X208			
	009	X209			
	010	X210			
	011	X211			
101CH (OUTPUT)	000	Y10100	UP	} GAUGE ROTATE VALVE	
	001	Y10101	DOWN		
	002	Y10102			
	003	Y10103			
	004	Y10104			
	005	Y10105			
	006	Y10106			
	007	Y10107			
	COM	0Vdc	GND		0V
	COM	0Vdc	GND		0V
102CH (OUTPUT)	000	Y10200			
	001	Y10201			
	002	Y10202			
	003	Y10203			
	004	Y10204			
	005	Y10205			
	006	Y10206			
	007	Y10207			
	COM	0Vdc	GND		0V
	COM	0Vdc	GND		0V

SMC ACTUATOR Wiring

Model	SMC Ref	Label	Description	Remark	Device Pin
LECPAN3D CN1	BK RLS				
	EMG	24Vdc	24V Power supply		24Vdc
	C 24V	24Vdc	24V Power supply		24Vdc
	M 24V	24Vdc	24V Power supply		24Vdc
	0V	0Vdc	GND		0Vdc
LECPAN3D CN5	001	24Vdc	24V Power supply		24Vdc
	002	0Vdc	GND		0Vdc
	003	5Vdc	NP+		5Vdc
	005	5Vdc	PP+		5Vdc

PLC I/O AND WIRING DIAGRAM TYPE VOLDEMORT

Model	PLC Mark	Label	Description	Remark	Device Pin
CP1E NA20DT-D	DC24V+	24Vdc	24V Power supply		24Vdc
	DC24 -	0Vdc	GND		GND
	E	E1	Earth		Earth
	COM	24Vdc	24V Power supply		24Vdc
CIO 0 (INPUT)	00	X000	START		
	01	X001	RESET		
	02	X002	EMERGENCY		
	03	X003	ACTUATOR ORIGIN		OMRON
	04	X004	PUSH BUTTON 1	} PART DETECT	
	05	X005	PUSH BUTTON 2		
	06	X006	PUSH BUTTON 3		
	07	X007	PUSH BUTTON 4		
	08	X008	SMC.BUSY (Actuator Moving)		SMC Driver 1.CN5.14
	09	X009	SMC.SETON (Returned to Origin)		SMC Driver 1.CN5.15
	10	X010	SMC.INP (Position Complete)		SMC Driver 1.CN5.16
	11	X011	SMC.ALARM		SMC Driver 1.CN5.19
CIO 90 (INPUT)		V IN0	ANALOG VOLTAGE INPUT		WIRE BLACK
		I IN0			
		COM 0	ANALOG VOLTAGE GROUND		SHIELD
		AG			
CIO 91 (INPUT)		V IN1			
		I IN1			
		COM 1			
CIO 100 (OUTPUT)	00	Y10000	Stepper Driver 1	CW - Add 390 Ohm resistor	SMC Driver 1.CN5.6
	01	Y10001			
	02	Y10002	Stepper Driver 1	CWW - Add 390Ohm resistor	SMC Driver 1.CN5.4
	03	Y10003			
	04	Y10004	SMC.SETUP (HOME)		SMC Driver 1.CN5.7
	05	Y10005	SMC.CLR(RESET DEVIATION)		SMC Driver 1.CN5.10
	06	Y10006	SMC.RESET (reset Alarm)		SMC Driver 1.CN5.8
	07	Y10007	SMC.SVON (Servo Online)		SMC Driver 1.CN5.9
	COM	0Vdc	GND		0V
	COM	0Vdc	GND		0V
	COM	0Vdc	GND		0V
	COM	0Vdc	GND		0V
CIO 190 (OUTPUT)		V OUT0			
		I OUT0			
		COM 0			

CP1W-40EDT					
1CH (INPUT)	000	X100	UP	} GAUGE ROTATE CYLINDER	
	001	X101	DOWN		
	002	X102			
	003	X103			
	004	X104			
	005	X105			
	006	X106			
	007	X107			
	008	X108			
	009	X109			
	010	X110			
	011	X111			
	COM	24Vdc	24V Power supply		24Vdc
2CH (INPUT)	000	X200	DUT SENSOR 1	} ORIGIN DUT MOTOR	OMRON
	001	X201	DUT SENSOR 2		OMRON
	002	X202	15MM GAP SENSOR	PROXIMITY	AUTONICS
	003	X203			
	004	X204			
	005	X205			
	006	X206			
	007	X207			
	008	X208			
	009	X209			
	010	X210			
	011	X211			
101CH (OUTPUT)	000	Y10100	UP	} GAUGE ROTATE VALVE	
	001	Y10101	DOWN		
	002	Y10102			
	003	Y10103			
	004	Y10104			
	005	Y10105			
	006	Y10106			
	007	Y10107			
	COM	0Vdc	GND		0V
	COM	0Vdc	GND		0V
102CH (OUTPUT)	000	Y10200			
	001	Y10201			
	002	Y10202			
	003	Y10203			
	004	Y10204			
	005	Y10205			
	006	Y10206			
	007	Y10207			
	COM	0Vdc	GND		0V
	COM	0Vdc	GND		0V

SMC ACTUATOR Wiring

Model	SMC Ref	Label	Description	Remark	Device Pin
LECPAN3D CN1	BK RLS				
	EMG	24Vdc	24V Power supply		24Vdc
	C 24V	24Vdc	24V Power supply		24Vdc
	M 24V	24Vdc	24V Power supply		24Vdc
	0V	0Vdc	GND		0Vdc
LECPAN3D CN5	001	24Vdc	24V Power supply		24Vdc
	002	0Vdc	GND		0Vdc
	003	5Vdc	NP+		5Vdc
	005	5Vdc	PP+		5Vdc

